

Kavya Annavarapu | Senior Data Analyst

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SUMMARY

Senior Data Analyst with 5+ years of experience designing scalable data solutions, optimizing SQL databases, and building ETL/ELT pipelines for enterprise analytics and regulatory reporting. Experienced in Python, PySpark, Snowflake, AWS, and Databricks to support cloud-based data engineering and large-scale data processing. Skilled in integrating LLMs, Retrieval-Augmented Generation (RAG), and machine learning into data pipelines for unstructured data processing and automated validation. Strong background in data modeling, data governance, and query performance optimization for financial analytics, risk management, and compliance.

SKILLS

Programming & Query Languages: SQL, T-SQL, Python, PySpark, Pandas, NumPy, Scikit-learn
Business Intelligence & Visualization: Power BI, Tableau, Dashboard Development, Executive Reporting, Interactive Reporting, KPI Reporting, Self-Service Analytics, Drill-Through Reporting
Data Analysis & Analytics: Business Analytics, Financial Analytics, Risk Analytics, Trend Analysis, Forecasting, Ad-hoc Analysis, Root Cause Analysis, Predictive Analytics, Anomaly Detection, Data Interpretation
Data Engineering & Integration: ETL, ELT, Data Ingestion, Data Transformation, Data Cleansing, Data Validation, Data Quality, Data Integration, Workflow Automation
Database & Query Optimization: Recursive CTEs, Window Functions, Stored Procedures, Views, Triggers, Query Performance Optimization, Execution Plan Analysis, Cross-Database Queries
Data Modeling & Warehousing: Logical Data Modeling, Physical Data Modeling, Data Warehousing, Snowflake, SQL Server, Schema Documentation, Metadata Management, Data Lineage
Cloud & Big Data Platforms: AWS, Amazon S3, AWS Glue, AWS Textract, Databricks, Apache Airflow
Artificial Intelligence & Machine Learning: Large Language Models (LLMs), LLM APIs, Retrieval-Augmented Generation (RAG), Machine Learning, OCR, Unstructured Data Processing, Metadata Classification, Schema Drift Detection, Vector Databases, Embedding Generation
Data Governance & Compliance: Data Governance, Regulatory Reporting, Financial Reporting, CCAR, Basel III, PII Masking, Data Integrity, Compliance Validation
Development Tools & Methodologies: GitHub Copilot, Amazon Q, Agile, Cross-Functional Collaboration, Stakeholder Management, Technical Mentoring

EXPERIENCE

Senior Data Analyst | Truist Financial | VA, USA Oct 2024 – Present

- Architected advanced SQL solutions using recursive CTEs, partition-based window functions, cross-database views, and execution plan analysis to optimize enterprise banking workloads, reducing query runtime by 40% and accelerating financial reporting.
- Implemented Python (PySpark, Pandas, NumPy) automation frameworks for multi-source data ingestion, ledger reconciliation, and data quality validation while leveraging GitHub Copilot and Amazon Q to streamline development, testing, and documentation.
- Designed scalable ETL/ELT pipelines and cloud-native workflows using AWS, Snowflake, and Apache Airflow to consolidate high-volume transactional data into governed data platforms, improving regulatory data readiness by 50%.
- Established logical and physical data models supporting CCAR, Basel III, and enterprise financial reporting while strengthening metadata management, governance, and validation across cloud environments.
- Integrated LLM APIs and machine learning models into analytics workflows to automate metadata classification, unstructured document processing, anomaly detection, schema drift monitoring, and compliance validation, reducing manual classification errors by 30%.
- Partnered with business stakeholders and Agile teams to deliver enterprise reporting solutions, optimize cloud analytics platforms, mentor junior analysts, and provide trusted datasets for risk and compliance initiatives.

Data Analyst | CapitalOne | VA, USA Jul 2022 – Sep 2024

- Created complex SQL queries using CTEs, window functions, joins, and stored procedures to analyze customer, credit, and financial datasets, improving reporting performance by 35% through query optimization.
- Produced interactive Power BI dashboards and executive reports tracking customer acquisition, portfolio health, financial KPIs, and risk metrics to support strategic decision-making.
- Streamlined Python (Pandas, NumPy, Scikit-learn) workflows for data cleansing, predictive anomaly detection, financial reconciliation, and API-based reporting, reducing manual auditing effort by 25%.
- Optimized Snowflake data models and ETL processes integrating cloud and on-premise data into centralized analytical repositories while maintaining data lineage, governance, and reporting accuracy.
- Performed ad-hoc analysis, trend analysis, and forecasting to identify operational inefficiencies, improve portfolio performance, and deliver actionable business recommendations.
- Collaborated with business stakeholders, data engineers, and product teams in Agile environments while utilizing GitHub Copilot and Amazon Q to accelerate SQL and Python development.

Junior Data Analyst | HCL | India Nov 2019 – Jul 2021

- Generated SQL queries, stored procedures, views, and triggers to extract, validate, and analyze operational data, improving query performance by 35% through indexing and execution plan optimization.
- Delivered Tableau dashboards and scheduled reports that monitored business KPIs, service performance, and operational trends for business stakeholders.
- Automated Python (Pandas, NumPy) workflows for data cleansing, anomaly detection, and report generation, reducing manual reporting effort by 40% while improving data accuracy.
- Maintained ETL processes migrating legacy data into SQL Server data warehouses while performing transformation, validation, and quality assurance for enterprise reporting.
- Supported logical and physical data modeling, metadata management, schema documentation, and governance initiatives to improve reporting consistency and enterprise data quality.
- Coordinated with DBAs, BI developers, and business users to resolve reporting issues, conduct root cause analysis, and deliver reliable datasets for operational and executive reporting.

PROJECTS

Executive Financial Performance Dashboard

Tech Stack: Power BI, SQL, Snowflake, Python, AWS

- Developed interactive Power BI dashboards visualizing loan performance, portfolio risk, customer acquisition, delinquency trends, and executive KPIs using enterprise financial datasets.
- Designed SQL data models and ETL workflows integrating Snowflake and AWS data sources into centralized reporting environments supporting self-service analytics.
- Automated Python data preparation and KPI calculations, reducing manual reporting effort by 40% while improving reporting accuracy.
- Implemented forecasting, trend analysis, and drill-through reporting capabilities enabling executives to identify performance drivers and business opportunities.

Intelligent Financial Document Analytics Platform

Tech Stack: Python, PySpark, AWS Textract, LangChain, Databricks, Amazon S3, Power BI

- Built an automated analytics pipeline extracting structured information from financial documents using OCR, AWS Textract, and distributed PySpark processing.
- Integrated LLM APIs and Retrieval-Augmented Generation (RAG) to classify financial documents, generate metadata, and support intelligent document search.
- Developed Power BI dashboards visualizing document processing performance, validation accuracy, compliance metrics, and operational KPIs.
- Implemented automated PII masking, data quality validation, metadata lineage, and anomaly detection, reducing manual document processing effort by 50% while maintaining regulatory compliance.

EDUCATION

Doctorate of education in computer science (Ongoing)

Judson University | IL, USA

Master of Science in Computer Science

Governors State University, University Park | IL, USA

Dec 2022

Bachelor of Technology in Computer Science

Jawaharlal Nehru Technological University | India

Oct 2020